

# picoCTF: Trivial Flag Transfer Protocol (TFTP)

**Category:** Forensics  
**Difficulty:** Medium  
**Challenge:** Figure out how they moved the flag  
**File:** tftp.pcapng

## What You Need

- Wireshark or tshark (for packet analysis)
- steghide (for steganography extraction)
- Basic understanding of ROT13 cipher

## Step 1: Extract Files from the PCAP

The pcap contains TFTP (Trivial File Transfer Protocol) traffic. TFTP is used to transfer files over UDP.

### Using Wireshark GUI:

```
File -> Export Objects -> TFTP -> Save All
```

### Using tshark command line:

```
tshark -r tftp.pcapng --export-objects tftp,./tftp_files
```

### Files Extracted:

| File             | Description                        |
|------------------|------------------------------------|
| instructions.txt | Encoded message (ROT13)            |
| plan             | Encoded message (ROT13)            |
| program.deb      | Debian package (steghide tool)     |
| picture1.bmp     | Image file                         |
| picture2.bmp     | Image file                         |
| picture3.bmp     | Image file (contains hidden flag!) |

## Step 2: Decode the ROT13 Messages

Both text files are encoded with ROT13 (Caesar cipher with shift of 13).

### Decode instructions.txt:

```
GSGCQBRFAGRAPELCGBHEGENSSVPFBJRZHFQGVFTHVFRBHESYNTGENAFSRE.SVTERBHGNJNLGBUVQRGURSYNTNAQVJVYYPURPXONPXSBEGURCYNA
```

### Decoded:

```
TFTP DOESNT ENCRYPT OUR TRAFFIC SO WE MUST DISGUISE OUR FLAG TRANSFER. FIGURE OUT A WAY TO HIDE THE FLAG  
↔ AND I WILL CHECK BACK FOR THE PLAN
```

### Decode plan:

```
VHFRQGURCEBTENZNAQUVQVGJVGU-QHRQVYVTRAPR.PURPXBHGGURCUBGBF
```

### Decoded:

```
I USED THE PROGRAM AND HID IT WITH-DUE DILIGENCE. CHECK OUT THE PHOTOS
```

### Key clues from decoded messages:

- They used “the program” (program.deb) to hide the flag
- Password hint: **DUE DILIGENCE**
- Check the photos (the .bmp files)

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## Step 3: Extract the Program

The program.deb is a Debian package. Let's extract it:

```
mkdir -p deb_extract  
dpkg-deb -x program.deb deb_extract/
```

Or install it:

```
sudo dpkg -i program.deb
```

**What it is:** The package installs steghide - a steganography tool that hides data in image files.

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## Step 4: Extract the Hidden Flag

The hint says they hid the flag “WITH-DUE DILIGENCE”. The password is **DUEDILIGENCE** (uppercase, no space).

**Important:** The flag is hidden in **picture3.bmp** (not picture1 or picture2!)

```
steghide extract -sf picture3.bmp -p DUEDILIGENCE
```

### Output:

```
wrote extracted data to "flag.txt"
```

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## Step 5: Read the Flag

```
cat flag.txt
```

Flag:

picoCTF{h1dd3n\_1n\_pLa1n\_51GHT\_18375919}

Complete Command Summary

```
# 1. Extract TFTP files from pcap
tshark -r tftp.pcapng --export-objects tftp,./tftp_files

# 2. Decode ROT13 messages (use online decoder or Python)
# instructions.txt and plan

# 3. Extract/install steghide from program.deb
mkdir -p deb_extract
dpkg-deb -x program.deb deb_extract/

# 4. Extract hidden flag from picture3.bmp
./deb_extract/usr/bin/steghide extract -sf picture3.bmp -p DUE DILIGENCE

# 5. Read flag
cat flag.txt
```

What Went Wrong (Common Mistakes)

| Mistake                 | Why It Failed              | Correct Approach              |
|-------------------------|----------------------------|-------------------------------|
| Used lowercase password | steghide is case-sensitive | Use DUE DILIGENCE (uppercase) |
| Tried picture1.bmp      | No data hidden there       | Use picture3.bmp              |
| Tried picture2.bmp      | No data hidden there       | Use picture3.bmp              |
| Password with space     | Wrong format               | DUE DILIGENCE (no space)      |

Key Takeaways

- 1. **TFTP transfers files** - always extract and check them
- 2. **Text files might be encoded** - try ROT13 for uppercase text
- 3. **.deb packages are installable software** - might be the tool you need
- 4. **steghide hides data in images** - needs password and correct image
- 5. **Read all hints carefully** - “DUE DILIGENCE” = password
- 6. **Try all files** - flag could be in any of them

Tools Used

| Tool             | Purpose                         |
|------------------|---------------------------------|
| Wireshark/tshark | Extract TFTP files from pcap    |
| ROT13 decoder    | Decode encrypted messages       |
| dpkg-deb         | Extract Debian packages         |
| steghide         | Extract hidden data from images |

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## Flag

```
picoCTF{h1dd3n_1n_pLa1n_51GHT_18375919}
```

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*Happy Forensics Hunting!* 🕵️